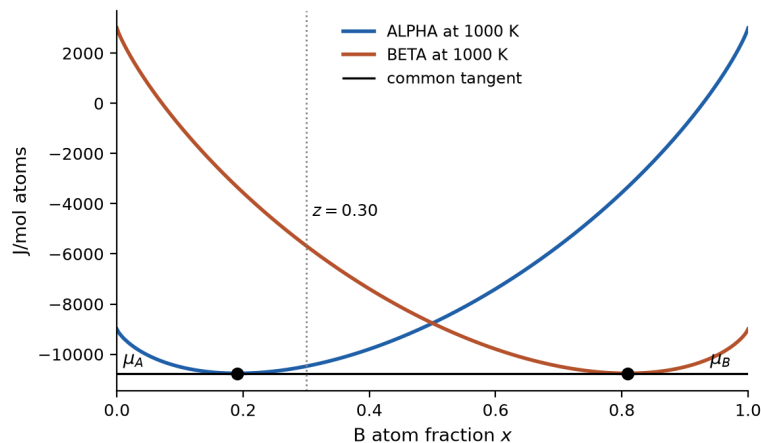


Two phases: a recap sheet

Use this sheet before Task 01 or the advanced steps 07–18. It puts four pictures together: the common tangent, the lever rule, the tie line with the melting lens, and one phase model that forms two phases. You need Day 2's D1 (counting A and B) and D3 (tangent end heights). Each part has one short check; the answers are at the end. All energies are J/mol atoms, and x and z are B atom fractions, as on Day 2. The longer version of this sheet is self-study step 03, parts A–C.

1. The common tangent



Day 2's ALPHA and a second phase model, BETA (its mirror image), at 1000 K. A sample whose overall B fraction z lies between the two touching points does better by splitting into ALPHA at $x = 0.191$ and BETA at $x = 0.809$ than by staying in either phase. One straight line touches both curves: the **common tangent**. As in D3, its end heights are μ_A and μ_B , and they are the same in both phases. In this mirror-image pair both equal -10762.7 , so the line is flat.

Check 1. At $z = 0.30$, which single phase is lower, and is the split lower still? Use $g_{\text{ALPHA}}(0.30) = -10479.0$, $g_{\text{BETA}}(0.30) = -5679.0$ and the tangent height -10762.7 .

2. The lever rule (you did this in D1 Q2)

The split must keep every atom: the amounts add to one, and the B atoms add to z :

$$f_{\text{ALPHA}} + f_{\text{BETA}} = 1, \quad 0.191 f_{\text{ALPHA}} + 0.809 f_{\text{BETA}} = z.$$

Solving gives the **lever rule**: $f_{\text{BETA}} = (z - 0.191)/(0.809 - 0.191)$. The phase nearer to z gets the larger amount.

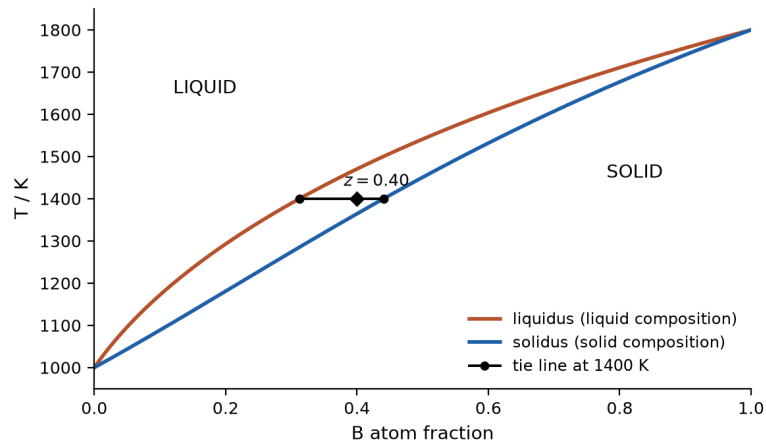
Check 2. Find f_{ALPHA} and f_{BETA} at $z = 0.30$, and check the B balance.

3. The chord rule

A split's energy is the amount-weighted sum of its two parts, $G = f_1 g_1 + f_2 g_2$. Drawn on the g - x plot, that point lies on the straight line (the **chord**) between the two dots, at $x = z$. Every split is a chord, so the lowest energy at z is the lowest chord, and the lowest chord is the common tangent.

Check 3. Why can no split of this sample lie below the common tangent at z ?

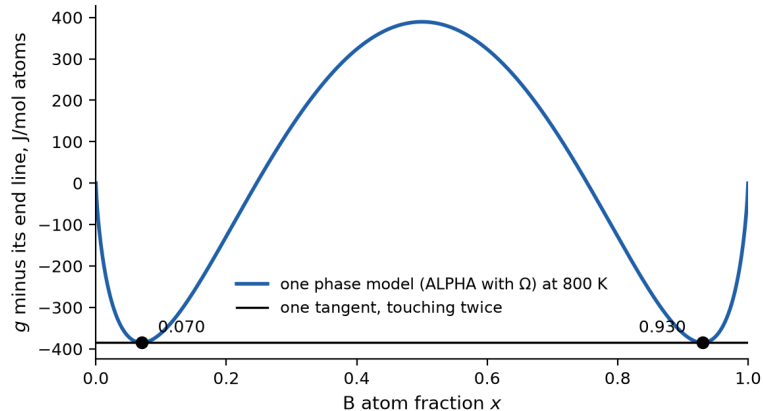
4. Tie line and lens



The melting lens of step 03 part C: an invented solid and liquid in which A melts at 1000 K and B at 1800 K. At 1400 K the common tangent touches the liquid at $x = 0.312$ and the solid at $x = 0.441$. On a temperature-composition diagram those two points are joined by a horizontal **tie line**. Repeating this at every temperature traces the two lens curves. A sample on the tie line splits: read the phase compositions at its ends, and the amounts with the lever rule along it.

Check 4. At 1400 K and $z = 0.40$, find f_{SOLID} .

5. One phase model, two phases



Day 2's ALPHA plus an energy cost for unlike neighbours, $\Omega x(1-x)$ with $\Omega = 20000$ J/mol, at 800 K (step 03 part B). The picture shows g minus its straight end line: the touching points are the same, and the hump is easier to see. One tangent touches the **same** curve twice, at 0.070 and 0.930. A sample in between splits into an A-rich and a B-rich region with the same crystal structure. This is a **miscibility gap**: one phase model, two phases.

Check 5. At $z = 0.20$, what fraction of the sample is the B-rich region?

Answers (after your attempt)

1. ALPHA is lower than BETA at $z = 0.30$ (-10479.0 against -5679.0). The split is lower still: -10762.7 , so about 284 J/mol atoms below ALPHA alone.
2. $f_{\text{BETA}} = (0.30 - 0.191)/(0.809 - 0.191) = 0.109/0.618 \approx 0.176$ and $f_{\text{ALPHA}} \approx 0.824$.
B balance: $0.824 \times 0.191 + 0.176 \times 0.809 \approx 0.157 + 0.142 = 0.30$.

3. Every split's energy lies on a chord between two points of the curves, and no part of either curve lies below the common tangent. So every chord, and every split, lies on or above it.
4. $f_{\text{SOLID}} = (0.40 - 0.312)/(0.441 - 0.312) = 0.088/0.129 \approx 0.68$, and the liquid is the remaining 0.32.
5. $(0.20 - 0.070)/(0.930 - 0.070) = 0.13/0.86 \approx 0.15$; the A-rich region is about 0.85.